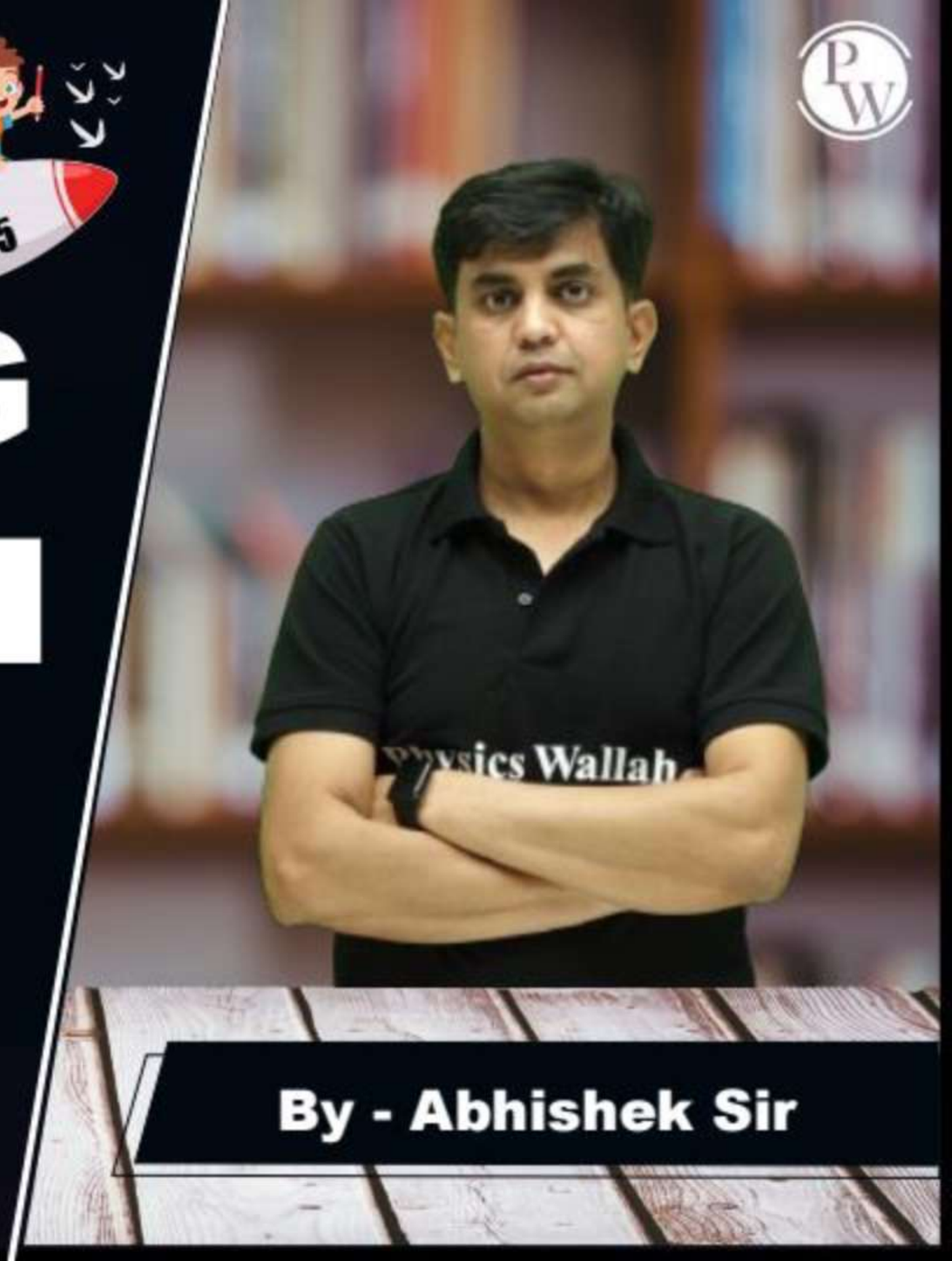
CS & IT ENGINEERING

Computer Network

Application Layer

Lecture No. - 03



By - Abhishek Sir

Recap of Previous Lecture

Topic DNS

Topic HTTP

Topics to be Covered

Topic HTTP

Topic Email

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW

@ABHISHEKSIRCS_PW

Topic : HTTP Connections

→ Two types of HTTP connections :

1. Non-persistent HTTP : HTTP/1.0
   → At most one object sent over TCP connection
   → Downloading multiple objects required multiple connections

2. Persistent HTTP : HTTP/1.1
   → Multiple objects can be sent over single TCP connection [between Client and Server]

Topic : Non-persistent HTTP (HTTP/1.0)

HTTP response time (per object) :

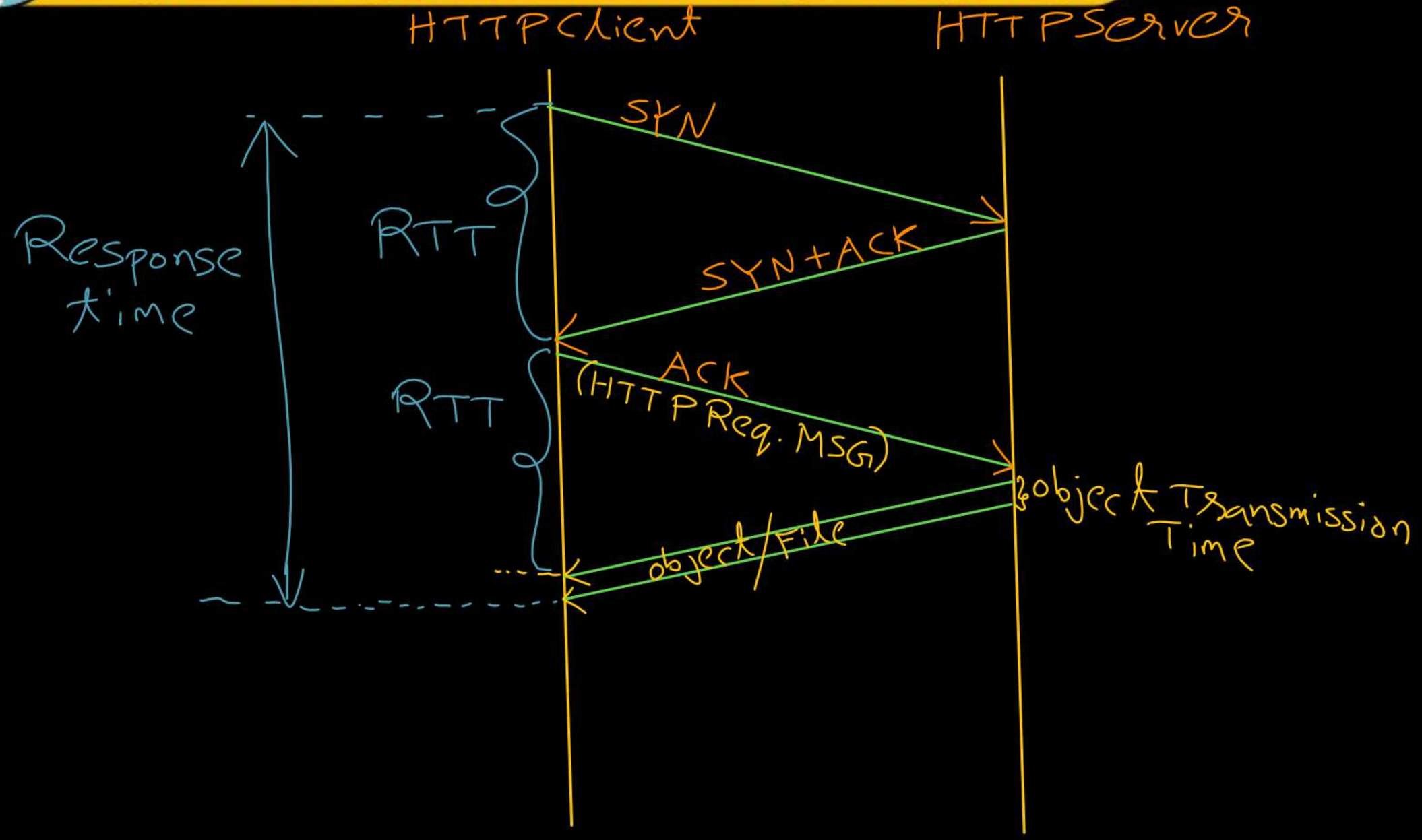
→ One RTT to initiate TCP connection

→ One RTT for HTTP request and first few bytes of HTTP response to return

→ Object / File transmission time

Response time = 2 * RTT + File Transmission Time

Topic : Non-persistent HTTP (HTTP/1.0)



#Q. Assume that you have made a request for a web page through your web browser to a web server. Initially the browser cache is empty. Further, the browser is configured to send HTTP requests in non-persistent mode. The web page contains text and five very small images. The minimum number of TCP connections required to display the web page completely in your browser is _____ .

[GATE-2020, 1-Mark]

IIT-D

Text

Image$_1$: Ref_URL$_1$
Image$_2$: Ref_URL$_2$
⋮
Image$_5$: Ref_URL$_5$

① TCP Conn. → Text File
② TCP Conn → Image$_1$
⋮
⑥ TCP Conn → Image$_5$

Ans = 6

No. of RTT in previous Question = ?

$Ans_1 = 6 * 2RTT = 12 RTT$

without parallel TCP Connection (default)

Ans_2 = with 5 parallel TCP Connection

① 2RTT ⇒ Text file

② 2RTT ⇒ All Five image object

$Ans_2 = 4 RTT$

Topic : Non-persistent HTTP (HTTP/1.0)

→ Requires 2 RTT per object

→ Operating System overhead for each TCP connection

→ Browser often open multiple parallel TCP connection to fetch referenced objects in parallel

Topic : Persistent HTTP (HTTP/1.1)

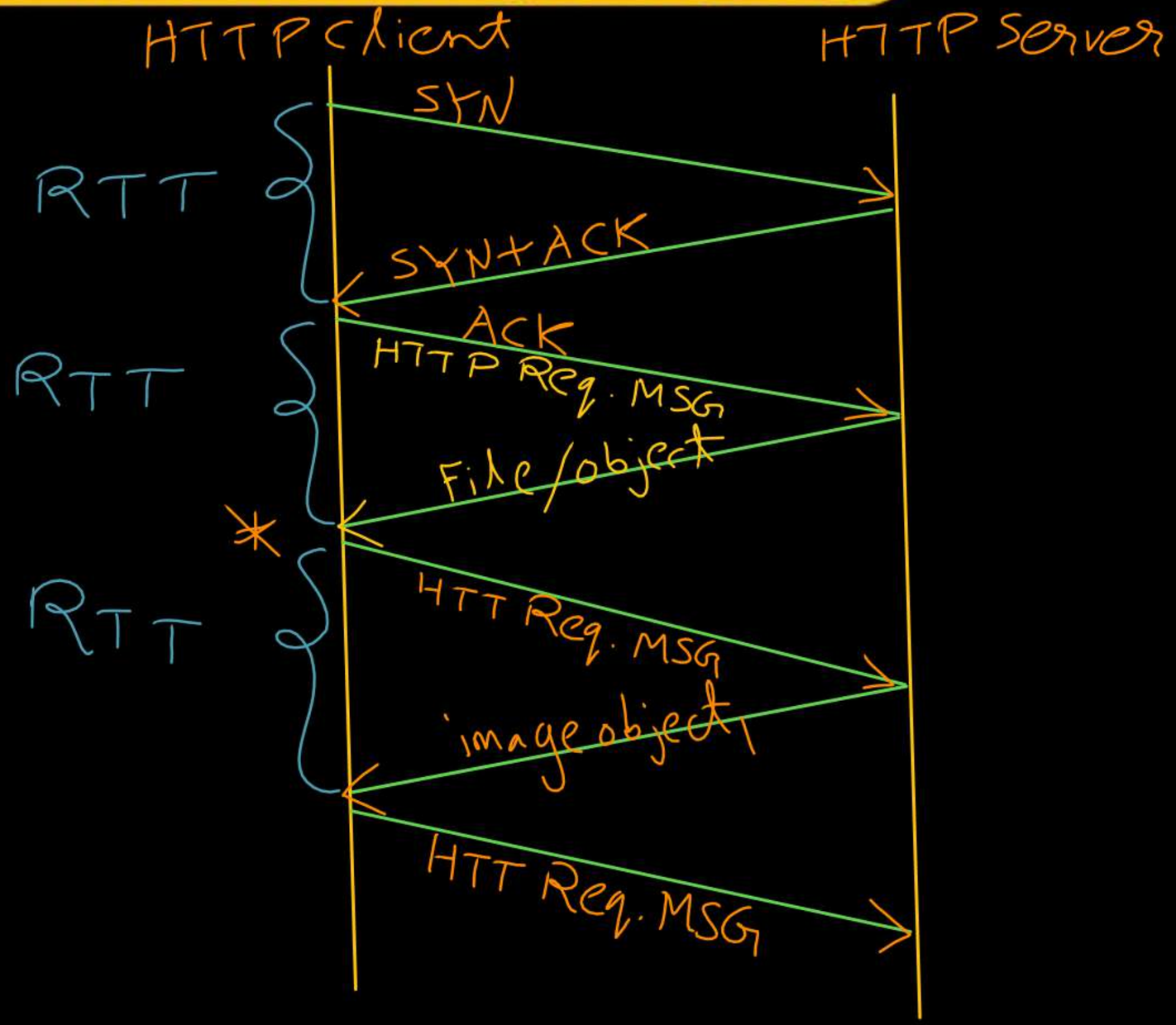
→ Server leaves connection open after sending response

→ Subsequent HTTP messages between same client/server sent over open connection

→ Client sends requests as soon as it encounters a referenced object

→ One RTT for each referenced objects (instead of two)

Topic : Persistent HTTP (HTTP/1.1)

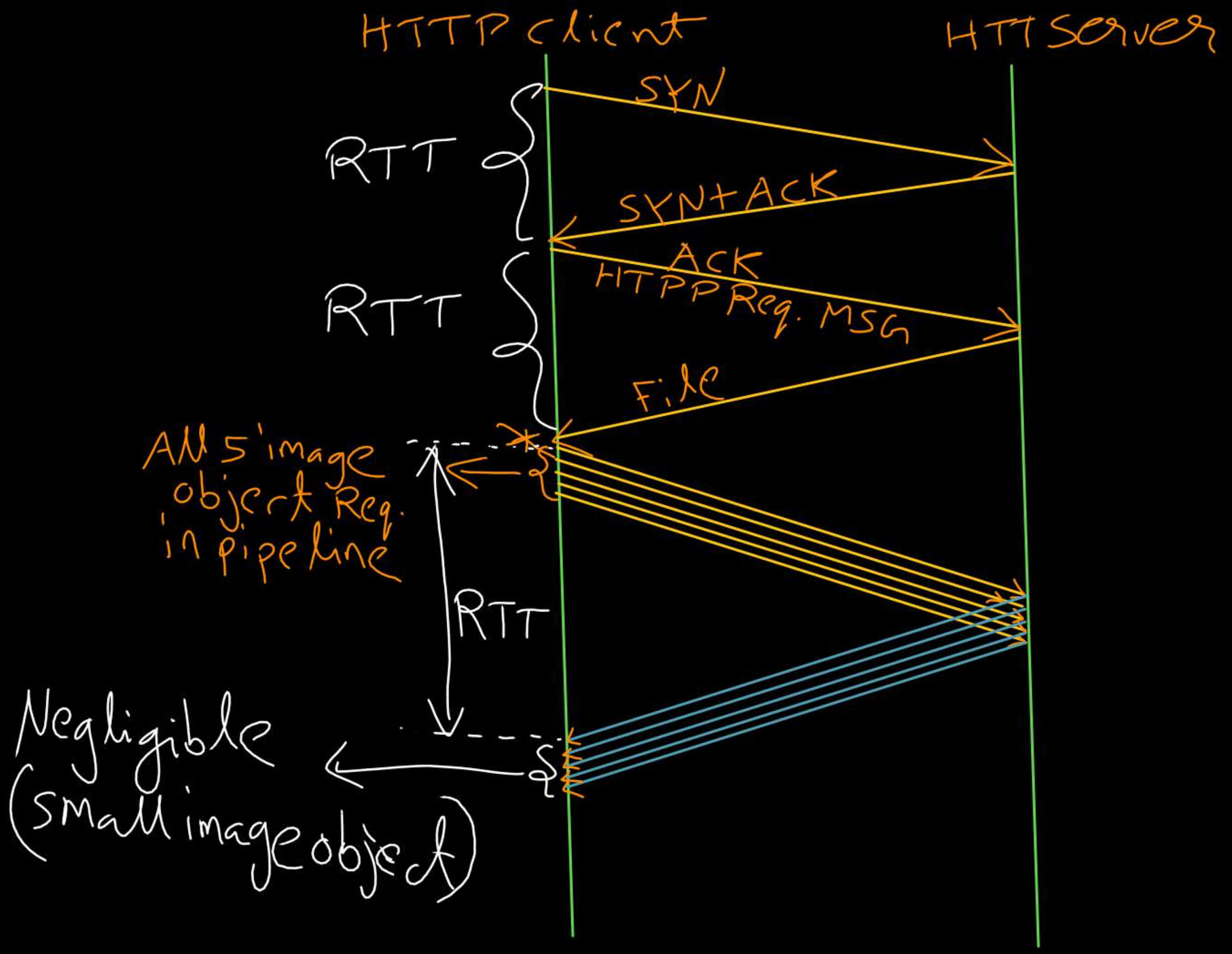


Persistent HTTP :-

one HTML/text file ✓
and five small image objects in same server

Ans_1 = Total 7 RTT (without pipeline) (default)

Ans_2 = Total 3 RTT (with pipeline)


HTTP client
HTT server
SYN
RTT
SYN+ACK
ACK
HTTP Req. MSG
RTT
File
All 5 image object Req. in pipeline
RTT
Negligible (small image object)

MCQ

#Q. A graphical HTML browser resident at a network client machine Q accesses a static HTML webpage from a HTTP server S. The static HTML page has exactly one static embedded image which is also at S. Assuming no caching, which one of the following is correct about the HTML webpage loading (including the embedded image)?

IIT KGP

[GATE-2014, Set-2, 2-Mark]

(A) Q needs to send at least 2 HTTP requests to S, each necessarily in a separate TCP connection to server S TRUE for Non-persistent HTTP

(B) Q needs to send at least 2 HTTP requests to S, but a single TCP connection to server S is sufficient TRUE for persistent HTTP

(C) A single HTTP request from Q to S is sufficient, and a single TCP connection between Q and S is necessary for this

(D) A single HTTP request from Q to S is sufficient, and this is possible without any TCP connection between Q and S

Ans: B

Topic : Persistent HTTP (HTTP/1.1)

→ Client can make requests in pipeline

→ Server can send referenced objects in pipeline

→ One RTT for all referenced objects (for small size objects)

IIT-K H.W. [MSQ]

#Q. Suppose in a web browser, you click on the www.gate-2023.in URL. The browser cache is empty. The IP address for this URL is not cached in your local host, so a DNS lookup is triggered (by the local DNS server deployed on your local host) over the 3-tier DNS hierarchy in an iterative mode. No resource records are cached anywhere across all DNS servers.

Let RTT denote the round trip time between your local host and DNS servers in the DNS hierarchy. The round trip time between the local host and the web server hosting www.gate-2023.in is also equal to RTT. The HTML file associated with the URL is small enough to have negligible transmission time and negligible rendering time by your web browser, which references 10 equally small objects on the same web server.

Which of the following statements is/are CORRECT about the minimum elapsed time between clicking on the URL and your browser fully rendering it?

[GATE-2023, 2-Mark]

(A) 7 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.

(B) 5 RTTs, in case of persistent HTTP with pipelining.

(C) 9 RTTs, in case of non-persistent HTTP with 5 parallel TCP connections.

(D) 6 RTTs, in case of persistent HTTP with pipelining.

#Q. Identify the correct sequence in which the following packets are transmitted on the network by a host when a browser requests a webpage from a remote server, assuming that the host has just been restarted.

[GATE-2016, Set-2, 1-Mark]

(MSC)

(A) HTTP GET request, DNS query, TCP SYN

(B) DNS query, HTTP GET request, TCP SYN

(C) DNS query, TCP SYN, HTTP GET request

(D) TCP SYN, DNS query, HTTP GET request

Ans: C

1. DNS Request/Query
   → DNS Reply
2. TCP SYN packet
   → TCP (SYN+ACK) packet
3. HTTP Request MSG

#Q. A user starts browsing a webpage hosted at a remote server. The browser opens a single TCP connection to fetch the entire webpage from the server. The webpage consists of a top-level index page with multiple embedded image objects. Assume that all caches (e.g., DNS cache, browser cache) are all initially empty. The following packets leave the user's computer in some order.

(i) HTTP GET request for the index page

(ii) DNS request to resolve the web server's name to its IP address

(iii) HTTP GET request for an image object

(iv) TCP SYN to open a connection to the web server

Which one of the following is the CORRECT chronological order (earliest in time to latest) of the packets leaving the computer ? (IISC) H.W

[GATE-2024, Set-1, 1-Mark]

(A) (iv), (ii), (iii), (i)

(B) (ii), (iv), (iii), (i)

(C) (ii), (iv), (i), (iii)

(D) (iv), (ii), (i), (iii)

Topic : HTTP Messages

→ Two types of HTTP Messages :

1. HTTP Request Message
2. HTTP Response Message

Topic : HTTP Request Message

→ ASCII (human-readable format)

→ HTTP request message contains :

1. Request Line
   → Method : GET / POST / PUT / HEAD
2. Header Lines
3. Body

Topic : HTTP Request Method

GET Method :

→ include user data in URL field of HTTP GET request message

POST Method :

→ Web page often includes form input

→ User input sent from client to server in entity body of HTTP POST request message

Topic : HTTP Request Method

PUT Method :

→ Upload new file (object) to server

→ Completely replaces file that exists at specified URL with content in entity body of HTTP PUT request message

HEAD Method :

→ Request headers (only)

→ if specified URL were requested with an HTTP GET Method

Topic : HTTP Response Message

→ HTTP response message contains :

1. Status Line
   → Contains 'Status code'
2. Header Lines
3. DATA

Topic : HTTP Response Status Code

→ Status code appear in first line in HTTP response message [sent from server to client]

200 : OK

→ Request succeeded, requested object later in this message

301 : Moved Permanently

→ Requested object moved, new location specified later in this message

Topic : HTTP Response Status Code

400 : Bad Request

→ Request message not understood by server

404 : Not Found

→ Request document not found on this server

505 :

→ HTTP version not supported

2 mins Summary

Topic HTTP

Topic [redacted]

THANK - YOU